BEING PSEUDO-COHERENT IS FPQC LOCAL ON THE BASE FOR MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17492-17496 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17498-17515, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pseudo-coherent twin of the collected perfect descent card L16092; same template, distinct property and distinct algebra input, counted per lane rules.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Descent, lemma descending-properties-morphisms; Definition definition-pseudo-coherent and chapter lemma flat-base-change-pseudo-coherent; More on Algebra, lemma flat-descent-pseudo-coherent.

Source reference. Original lemma 067Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17375–17381. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02KP: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17515–17515. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0680: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17388–17390. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 068R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17515–17515. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is pseudo-coherent" is fpqc local on the base.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. We will use the criterion of Descent, Lemma 02KP to prove this. By Definition 067Z being pseudo-coherent is Zariski local on the base. By Lemma 0680 being pseudo-coherent is preserved under flat base change. The final hypothesis (3) of Descent, Lemma 02KP translates into the following algebra statement: Let $A \to B$ be a faithfully flat ring map. Let $C = A[x_1, \ldots, x_n]/I$ be an A-algebra. If $C \otimes_A B$ is pseudo-coherent as an $B[x_1, \ldots, x_n]$-module, then C is pseudo-coherent as a $A[x_1, \ldots, x_n]$-module. This is More on Algebra, Lemma 068R. □